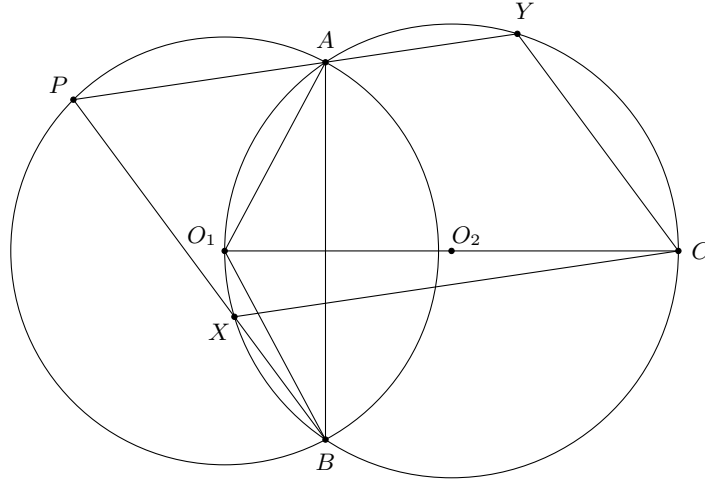




HK IMO Training 2019-2020
Problem Set 4 Suggested Solutions



Problem 13. Two circles ω_1 and ω_2 with centers O_1 and O_2 respectively intersect each other at points A and B , and point O_1 lies on ω_2 . Let P be an arbitrary point lying on ω_1 . Lines BP , AP and O_1O_2 cut ω_2 for the second time at points X , Y and C , respectively. Prove that quadrilateral $XPYC$ is a parallelogram.



Solution. We use directed angles to handle all cases together. Note that CO_1 is the perpendicular bisector of AB . This yields

$$\angle CO_1A = \frac{\pi}{2} - \angle O_1AB = \angle BPA.$$

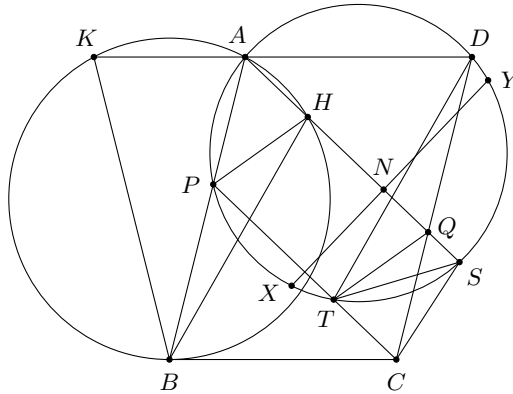
From the concyclic points, we have $\angle CO_1A = \angle CYA$. Therefore, we obtain $\angle BPA = \angle CYA$, which implies PX and YC are parallel. Next, we have

$$\angle CXP = \angle CXB = \angle CO_1B = \angle AO_1C = \angle AYC = \angle YPX$$

using the parallel lines obtained. This implies PY and XC are parallel.

Therefore, $XPYC$ is a parallelogram.

Problem 14. Let $ABCD$ be a parallelogram and let K be a point on line AD such that $BK = AB$. Suppose that P is an arbitrary point on AB , and the perpendicular bisector of PC intersects the circumcircle of $\triangle APD$ at points X, Y . Prove that the circumcircle of $\triangle ABK$ passes through the orthocentre of $\triangle AXY$.



Solution. Let AN be the altitude from A of $\triangle AXY$. Suppose that the circumcircle of $\triangle ABK$ intersects AN at the point H . It suffices to show that H is the orthocentre of $\triangle AXY$.

Suppose that PC and AN intersect the circumcircle of $\triangle APD$ for the second time at points T and S , respectively and AN intersects CD at Q . In the above configuration, we have

$$\angle BHS = \angle AKB = \angle KAB = \angle PTD.$$

Notice that AN and PC are parallel, since both are perpendicular to XY . Also, AP and CQ are parallel, so $APCQ$ is a parallelogram. This yields $\angle BAH = \angle DCT$. Also, we have

$$\angle AHB = 180^\circ - \angle BHS = 180^\circ - \angle PTD = \angle CTD$$

and $BA = DC$. These imply $\triangle AHB \cong \triangle CTD$, and hence $AH = CT$. So $PTQH$ is also a parallelogram. As $APTS$ is an isosceles trapezoid, we find that $CQ = AP = TS$. Therefore $TQSC$ is also an isosceles trapezoid. Finally, from $CS = TQ = PH$, $CSHP$ is an isosceles trapezoid.

Therefore, S and H are symmetric with respect to XY . So H must be the orthocentre of $\triangle AXY$. The proofs for other configurations are the same.

Problem 15. Consider an 8×8 chessboard with all 64 unit squares coloured white. We colour 12 unit squares black in an arbitrary manner. Prove that we can always find four rows and four columns containing the 12 black unit squares.

Solution. We first choose 4 rows that contain the largest number of black squares. We claim that there are at most 4 black squares not chosen. Suppose on the contrary that there are at least 5 black squares which are not chosen. Then at least one row which is not chosen contains $\left\lceil \frac{5}{4} \right\rceil = 2$

black squares. Also, there are at most $12 - 5 = 7$ black squares in the chosen rows, and one of them contains at most $\left\lfloor \frac{7}{4} \right\rfloor = 1$ black square. This contradicts the choice of the rows as we should have chosen the row with at least 2 black squares instead.

Therefore, at most 4 black squares lie in the rows which are not chosen. Then we can clearly choose 4 columns that contain all these black squares.

Problem 16. Determine whether there exists a finite set S of prime numbers such that, for every integer $n \geq 2$, the number $2^2 + 3^2 + \dots + n^2$ is divisible by some prime number $p \in S$.

Solution. Suppose S is such a set. Note that

$$2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} - 1 = \frac{2n^3 + 3n^2 + n - 6}{6}.$$

For each prime $p > 3$, the equation $2n^3 + 3n^2 + n - 6 \equiv 0 \pmod{p}$ has at most 3 solutions modulo p . Therefore, we can choose an integer a_p such that $2a_p^3 + 3a_p^2 + a_p - 6 \not\equiv 0 \pmod{p}$. Then we take any positive integer $n > 1$ such that

$$n \equiv 3 \pmod{4},$$

$$n \equiv 2 \pmod{9},$$

$$n \equiv a_p \pmod{p},$$

for any $p \in S$. Such an n exists by the Chinese remainder theorem. It is easy to check that $2^2 + 3^2 + \dots + n^2$ is odd and is congruent to 1 modulo 3. Thus, this sum is not divisible by 2 or 3 or any prime $p > 3$ in S . This is a contradiction. Therefore, no such S exists.